

Mathemalchemy Class Activity

Course _____ Date _____ Name _____

Mathemalchemy is a travelling mathematical art installation built by 24 artists and mathematicians. Answer from what you see there; mathemalchemy.org has more.

Part 1. Codes

1. The black and white squares on the skateboarder’s flag in the *Silhouettes* scene are a hidden message in binary, written in a code called **ASCII**. What is ASCII, and what does it have to do with the texts we all send every day?

2. What is **cryptography**? Why are **prime numbers** so important to it? In the Garden, what two numbers are the chipmunks sorting? What are the squirrels doing?

3. The Cryptography Quilt has 25 panels. Choose **five**. For each, give the cipher or idea it shows, and what it is used for (or was, historically).

Cipher or idea	What it is (or was) used for

Part 2. Infinite series

4. In mathematics, what do the words **converge** and **diverge** mean?

converge: _____

diverge: _____

5. The two Ball Arches rise from the very same first ball. What do we mean when we say the short arch **converges** and the long arch **diverges**?

6. The long arch is infinitely long, yet the total sum of the **volume** of the spheres converges. Explain what this means.

7. Only some of the balls on the arches are embroidered with designs. Which ones? Count the positions along the arch. Why do you think the artists chose those?

8. Does the infinite sum $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$ converge or diverge? If it converges, to what? Where does this sum appear in the exhibit?

9. **Real money, forever.** A **perpetuity** pays money forever, like an endowed scholarship. Say one pays **\$1,000 every year, forever**. A dollar you'll receive years from now is worth less than a dollar in your hand today; what a future payment is worth today is its **present value**. Those infinitely many \$1,000 payments have a *finite* total present value, a price you could actually pay today. Why? Which arch does a perpetuity behave like?

Part 3. Polyhedra

10. Name the five **Platonic solids**. Find at least one in the exhibit and say where it is.

11. What is **Euler's formula** for polyhedra? What are **polyhedra**?

12. Count the vertices, edges and faces of three Platonic solids, and check Euler's formula on each.

	vertices V	edges E	faces F	$V - E + F$
tetrahedron				
cube				
dodecahedron				

Now look up at the Lighthouse's dodecahedron. On a cube, a straight path from a corner always runs into another corner. The dodecahedron is the only Platonic solid where a straight path can leave a corner and return to it, touching no other corner. Proved in 2020.